AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MT101
General Mathematics

Final Examination
Fall Semester 2025/2026

Date: 15/Jan. /2026

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed: 2:00 Hours**

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Instructions:

- Total Marks: **50 marks**
- This exam consists of 5 questions
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Scientific Calculators** are allowed.

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You should answer all the questions, giving 50 possible marks. Show your work.

Q1. [2+3+2+ 3 Marks]

a) Given the function f(x) = $2x^2 + 8x$.

i) Find the domain of the function f.

ii) Find the vertex of the graph of the function f.

iii) Find the zeros for the quadratic function f(x).

b) Use synthetic division to determine if x+ 3 is a factor of *the function* $f(x) = x^3 + 3x^2 - 2x - 6$.

Q2. [3+4+3 marks]

a) Find the equation of the line passes through the point (2, 5) and has slope -2.

c) Find the inverse of the function f(x)= $\sqrt[3]{2x-6}$

c) The time required to travel a given distance is inversely proportional to the speed of travel. If a trip can be made in 3.6 h at a speed of 70 km/h, how long will it take to make the same trip at 90 km/h?

Q3. [4+2+4marks]

Solve the following inequality:

a) $x^2 - 5x + 4 < 0$

b) Find the domain of the following and write the answer in interval notation

i) F(x)= $\ln 3x$

ii) G(x)= $\sqrt{x-1}$

Q4: [3+3+4 marks]

Solve for x:

a) $3^{2x} = 5$

b) $\log_3(x) + \log_3(x-6) = 3$

c) $\log_a(x) = 2\log_a 3 + \log_a 5$

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Q5. [2+1+2+3+2marks]

a) You can order a hamburger with cheese, onion, pickle, relish, mustard, lettuce, tomato, or mayonnaise. How many different combinations of extras can you order, choosing any four of them?

b) Given $P(A)=\frac{1}{2}$; $P(B)=\frac{1}{5}$; and $p(A \cap B)=\frac{1}{10}$, Find

i) $P(\bar{B})$

ii) $P(A \cup B)$

iii) $P(B \mid A)$

iv) **Are the two events A and B independent? Justify your answer.**

End of questions

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